CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

Associativity and the co-smash product

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Abstract

The aim of this talk is to sketch some ideas in joint work-in-progress with Ülo Reimaa and Corentin Vienne. The original question we set out to answer was to characterise when the objects in a variety of algebras over a field K have a multiplication that is associative. What we found is a categorical description of the variety of commutative associative algebras over K amongst all so-called operadic varieties over K, where a set of multilinear identities suffices to describe the algebras in the variety. As it turns out, the variety of commutative associative algebras over K is the only operadic variety whose co-smash products are not just symmetric, but also associative in the sense of Carboni and Janelidze.